



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

Nitte (DU) established under Section 3 of UGC Act 1956 | Accredited with 'A+' Grade by NAAC

QUESTION BANK

Subject: Basic Electronics

Course Code: EC1001-1 (3 credits)

UNIT III

- 1 a. Explain the classification of an Embedded System.
- b. What is an Embedded System? Compare the difference between the Embedded and general purpose computing system.
- c. What is difference between RISC and CISC processors? Give an example for each.

- 2 a. Compare
 - i) Harvard and Von-Neumann architectures, with diagrams.
 - ii) Microprocessor and Microcontroller
- b. Explain the elements of an embedded system with block diagram.
- c. What is sensor and actuator? Explain how the LED and opto coupler are used as an I/O subsystem in embedded system.

3. With neat diagrams discuss the basic elements of communication systems.
4. Discuss the types of communication channels used for the transmission of signals.
5. Briefly discuss about the noise in communication systems.
6. Briefly discuss about the modulation process and also discuss types of modulation.
7. a) Explain the following schemes of cellular communication,
 - (i) Forward Control Channel
 - (ii) Reverse control channel

(iii) Duplexer

(iv) Hand-off

b) Draw the diagram of a cellular system and show the frequency reuse mechanism in it.